# Passively Damped LC Output Filter with R–C Damping Branch

$L$–$C_1$ filter with a $C_2$–$R_2$ damping branch at the output of an IGBT full bridge: theory, benefits and design criteria

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**Abstract**

This note deals with the second-order $L$–$C_1$ low-pass filter whose capacitor is shunted by a series $C_2$–$R_2$ branch, placed between an IGBT full bridge with unipolar PWM (switching frequency 1–10 kHz per leg, ripple at 2–20 kHz) and a load that absorbs DC or slow AC (up to about 5 Hz). The transfer functions, the asymptotic behaviour and the benefit of the damping branch with respect to the undamped filter and to the other passive damping options are derived. The classical single-resistor optimum of Erickson for the output impedance is recalled, and two further closed-form choices of $R_2$ (minimum peak of the voltage transfer function and pole placement with a real pole at $\omega_0$) are derived and compared. A step-by-step design procedure and a worked example with the loss budget of the damping resistor at the ripple frequency and at the slow-AC output frequency are given, together with a MATLAB script that reproduces the calculations.

Table 1: Revision history

| Rev. | Date | Description |
|---|---|---|
| 00 | 2026-09-21 | First issue. |

## Contents

**1 Introduction and scope** **2**
1.1 Application data assumed in this note . . . 2

**2 The filter and its transfer functions** **3**
2.1 Circuit . . . 3
2.2 Impedance of the shunt network . . . 3
2.3 Transfer functions . . . 4
2.4 Asymptotic behaviour . . . 5

**3 Why damping is needed and what the branch buys** **5**
3.1 The undamped filter with a current-source-like load . . . 5
3.2 Sources of excitation of the resonance . . . 5
3.3 Passive damping alternatives . . . 6

**4 Design theory of the damping branch** **6**
4.1 The point independent of $R_2$ . . . 6
4.2 Optimum for the output impedance (Erickson) . . . 8
4.3 Optimum for the voltage transfer function . . . 8
4.4 Pole placement: real pole at $\omega_0$ . . . 8

4.5 Summary table and choice of $n$ . . . 9
4.6 Sensitivity to the value of $R_2$ . . . 9
4.7 Time-domain response . . . 10

**5 Design criteria for the output filter of a full bridge 10**
5.1 Ripple source: unipolar PWM . . . 10
5.2 Design procedure . . . 11
5.3 Losses in the damping resistor and component ratings . . . 12

**6 Worked example 13**

**7 Practical notes 14**

**8 Conclusions 15**

**A Derivations 15**
A.1 Shunt network and transfer functions . . . 15
A.2 Common point of the magnitude curves . . . 16
A.3 Optimum values of $r$ . . . 16
A.4 Pole placement . . . 16
A.5 Unipolar PWM pulse train . . . 17

**B MATLAB script 17**

# 1 Introduction and scope

A full bridge of IGBTs operated with unipolar voltage switching produces at its output terminals a pulse train whose fundamental ripple lies at twice the switching frequency of each leg. When the load must receive a clean DC or slow-AC voltage (magnet coils, test loads, electrolytic or resistive processes, downstream regulated converters), a low-pass filter is inserted between the bridge and the load. The simplest filter is a series inductor $L$ and a shunt capacitor $C_1$: it attenuates the ripple at $-40\,\mathrm{dB/decade}$ above its resonance, but its resonance is practically undamped whenever the load does not behave as a resistor close to the characteristic impedance $\sqrt{L/C_1}$. This is exactly the situation with an inductive (current-source-like) load, with a light load, or with a tightly regulated downstream converter.

The topology studied here adds a series branch $C_2$–$R_2$ in parallel with $C_1$ (Fig. 1). The capacitor $C_2$ blocks the DC component so that $R_2$ does not dissipate the load power, while around the resonance $C_2$ is nearly a short and $R_2$ appears in parallel with $C_1$, damping the resonance. At high frequency $C_1$ dominates again and the $-40\,\mathrm{dB/decade}$ asymptote is preserved. This arrangement is known in the switching-converter literature as $R_f$–$C_b$ *parallel damping* of an input filter [2, 3, 4, 5] and as *RC passive damping* of LCL grid filters [11, 12, 13]; the analysis is the same for an output filter.

The document is organised as follows. Section 2 derives the transfer functions and the normalised parameters. Section 3 explains why damping is necessary in the application considered and compares the passive damping alternatives. Section 4 gives the design theory of the branch (choice of $C_2$ and $R_2$) with closed-form results. Section 5 turns this into a design procedure for the output filter of a full bridge, Section 6 works a numerical example and Section 7 collects practical notes. Appendix A contains the derivations and Appendix B the MATLAB script.

## 1.1 Application data assumed in this note

The application considered is an IGBT full bridge (H-bridge) supplied by a DC link $V_{dc}$, modulated with unipolar voltage switching (the two legs are driven by complementary references, so the

output pulses have amplitude $V_{dc}$ and frequency $f_r = 2f_{sw}$), with $f_{sw}$ between 1 kHz and 10 kHz. The output voltage is DC or a slow AC up to $f_{ac} \approx 5$ Hz; the load is essentially inductive (coil) with a series resistance, or a regulated converter. The worst case for the filter design is the lowest switching frequency, i.e. a ripple frequency $f_{r,\min} = 2$ kHz.

Table 2: Main symbols

| Symbol | Meaning |
|---|---|
| $L$, $C_1$ | main filter inductor and capacitor |
| $C_2$, $R_2$ | damping branch (series), in parallel with $C_1$ |
| $\omega_0 = 1/\sqrt{LC_1}$, $f_0 = \omega_0/2\pi$ | resonance of the undamped $L$–$C_1$ filter |
| $R_0 = \sqrt{L/C_1}$ | characteristic impedance of the $L$–$C_1$ filter |
| $n = C_2/C_1$ | capacitance ratio of the damping branch |
| $r = R_2/R_0$ | normalised damping resistance |
| $\Omega = \omega/\omega_0$, $p = s/\omega_0$ | normalised frequency and Laplace variable |
| $v_b$ | bridge output voltage (PWM), input of the filter |
| $v_o$, $i_o$ | filter output voltage and load current |
| $H(s) = v_o/v_b$ | voltage transfer function (unloaded) |
| $Z_o(s)$ | output impedance of the filter (source shorted) |
| $Z_{in}(s)$ | input impedance seen by the bridge (load open) |
| $f_{sw}$, $f_r = 2f_{sw}$ | leg switching frequency, output ripple frequency |
| $m$ | modulation index, $\overline{v_b} = m\,V_{dc}$, $-1 \le m \le 1$ |

# 2 The filter and its transfer functions

## 2.1 Circuit

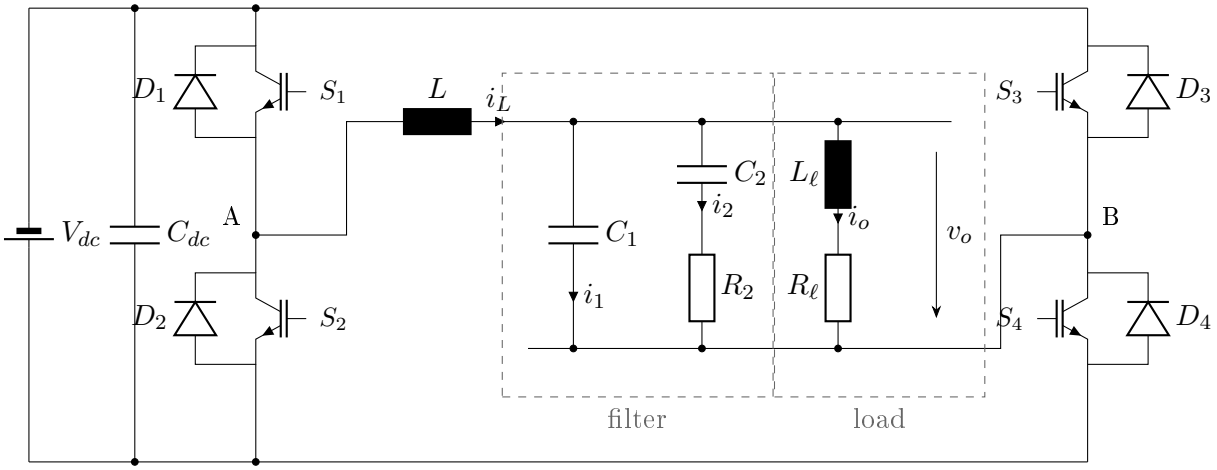



Figure 1: IGBT full bridge with the $L$–$C_1$ output filter and the $C_2$–$R_2$ damping branch; the load is represented by $L_\ell$–$R_\ell$ (magnet-type load). The bridge output voltage is $v_b = v_{AB}$.

## 2.2 Impedance of the shunt network

The shunt network is the parallel of $C_1$ and of the series branch $C_2$–$R_2$:

$$Z_c(s) = \frac{1}{sC_1} \parallel \left(R_2 + \frac{1}{sC_2}\right) = \frac{1 + sR_2C_2}{s\,(C_1 + C_2)\,\big(1 + sR_2C_s\big)}, \qquad C_s = \frac{C_1C_2}{C_1 + C_2}. \tag{1}$$

It has a zero at $\omega_z = 1/(R_2C_2)$ and a pole at $\omega_p = 1/(R_2C_s)$, with $\omega_p/\omega_z = 1+n$. Three frequency regions follow:

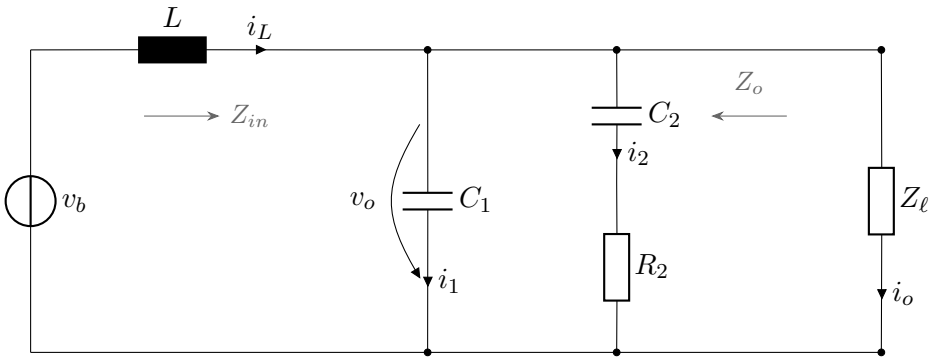



Figure 2: Equivalent circuit for the analysis. $Z_{in}$ is seen from the bridge with the load open; $Z_o$ is seen from the load with the bridge replaced by a short circuit.

- $\omega \ll \omega_z$: $C_2$ is an open circuit compared with $R_2$, and $Z_c \simeq 1/[s(C_1 + C_2)]$ – the two capacitors add;
- $\omega_z \ll \omega \ll \omega_p$: $Z_c \simeq R_2\, n/(1+n)$ – the network is resistive; this is the band in which the resonance must fall;
- $\omega \gg \omega_p$: $Z_c \simeq 1/(sC_1)$ – $R_2$ is large compared with the reactance of $C_1$ and the branch no longer matters.

In normalised form the corners are $\Omega_z = 1/(nr)$ and $\Omega_p = (1+n)/(nr)$. For the design values recommended later ($n = 2$–$4$, $r \approx 0.75$–$1$) one gets $\Omega_z \approx 0.33$–$0.5$ and $\Omega_p \approx 1.5$–$1.7$, so that the resonance region is indeed inside the resistive plateau.

### 2.3 Transfer functions

With the load open, the voltage transfer function, the output impedance and the input impedance are

$$H(s) = \frac{v_o}{v_b} = \frac{Z_c}{sL + Z_c} = \frac{1 + sR_2C_2}{1 + sR_2C_2 + s^2L(C_1+C_2) + s^3LR_2C_1C_2}, \tag{2}$$

$$Z_o(s) = sL \parallel Z_c = sL\,H(s) = \frac{sL\,(1 + sR_2C_2)}{1 + sR_2C_2 + s^2L(C_1+C_2) + s^3LR_2C_1C_2}, \tag{3}$$

$$Z_{in}(s) = sL + Z_c = \frac{1 + sR_2C_2 + s^2L(C_1+C_2) + s^3LR_2C_1C_2}{s(C_1+C_2)(1+sR_2C_s)}. \tag{4}$$

The filter is of third order. The relation $Z_o = sL\,H$ is exact and is used below: the two quantities have the same denominator, the same zero and differ only by the factor $sL$. With a load $Z_\ell$ the loaded transfer function is obtained by replacing $Z_c$ with $Z_c \parallel Z_\ell$ in (2).

Introducing $\omega_0$, $R_0$, $n$, $r$ and $p = s/\omega_0$, the denominator becomes

$$D(p) = 1 + nr\,p + (1+n)\,p^2 + nr\,p^3, \tag{5}$$

and

$$H(p) = \frac{1 + nr\,p}{D(p)}, \qquad \frac{Z_o(p)}{R_0} = \frac{p\,(1+nr\,p)}{D(p)}, \qquad \frac{Z_{in}(p)}{R_0} = \frac{D(p)}{p\,(1+n)\big(1+\frac{nr}{1+n}p\big)}. \tag{6}$$

For $p = j\Omega$ the magnitudes are

$$|H(j\Omega)| = \sqrt{\frac{1 + (nr\Omega)^2}{\big[1-(1+n)\Omega^2\big]^2 + \big[nr\Omega(1-\Omega^2)\big]^2}}, \qquad \frac{|Z_o(j\Omega)|}{R_0} = \Omega\,|H(j\Omega)|. \tag{7}$$

Everything therefore depends on the two dimensionless numbers $n$ and $r$ only; the absolute values $L$, $C_1$ enter through $\omega_0$ and $R_0$.

### 2.4 Asymptotic behaviour

The limiting cases of (7) are instructive:

- $r \to 0$ ($R_2$ shorted): $H = 1/[1-(1+n)\Omega^2]$, an undamped $L$–$(C_1+C_2)$ filter resonating at $\Omega = 1/\sqrt{1+n}$, i.e. $f_{low} = f_0/\sqrt{1+n}$;
- $r \to \infty$ ($R_2$ open): $H = 1/(1-\Omega^2)$, the undamped $L$–$C_1$ filter resonating at $\Omega = 1$;
- $\Omega \gg \Omega_p$: $|H| \simeq 1/\Omega^2 = (f_0/f)^2$, i.e. the $-40\,\mathrm{dB}$/decade asymptote of the $L$–$C_1$ pair, independent of the branch;
- $\Omega \ll \Omega_z$: $|H| \simeq 1 + (1+n)\Omega^2$, unity gain with negligible phase lag.

The branch therefore affects the response only in the decade around the resonance, which is precisely what a damping network should do. The price is the shift of the low-frequency corner from $f_0$ to $f_0/\sqrt{1+n}$: the filter starts to deviate from unity gain somewhat earlier than the bare $L$–$C_1$ pair.

## 3 Why damping is needed and what the branch buys

### 3.1 The undamped filter with a current-source-like load

With the branch removed, the quality factor of the $L$–$C_1$ resonance is set by the parasitic series resistance $R_L$ of the inductor (plus the ESR of $C_1$) and by the equivalent parallel resistance $R_p$ of the load at $f_0$:

$$Q = \frac{1}{\dfrac{R_L}{R_0} + \dfrac{R_0}{R_p}}. \tag{8}$$

A resistive load equal to $R_0$ would give $Q \approx 1$ and no damping problem. A magnet load $L_\ell$–$R_\ell$ with $\omega_0 L_\ell \gg R_0$ has a parallel equivalent $R_p = (R_\ell^2 + \omega_0^2 L_\ell^2)/R_\ell$ of hundreds or thousands of ohms and contributes nothing; the same holds at light load. With $R_L$ of a few milliohms and $R_0$ around one ohm, $Q$ of the order of 100–300 results. The consequences are:

- the peak of $|H|$ equals $Q$ (40–50 dB): any component of $v_b$ near $f_0$ is amplified by that factor;
- a step of the average bridge voltage produces a 100 % overshoot of $v_o$ that rings at $f_0$ with time constant $2Q/\omega_0$, i.e. hundreds of milliseconds;
- the peak of $|Z_o|$ equals $QR_0$: a load current step $\Delta i_o$ produces a ringing voltage of amplitude $\approx R_0 \Delta i_o$ that again decays with $2Q/\omega_0$;
- the current regulator sees a plant with a +40–50 dB resonance; the loop gain must be pushed well below 0 dB at $f_0$, which forces a crossover frequency far below $f_0$ or a notch that is sensitive to component tolerances.

### 3.2 Sources of excitation of the resonance

In the application considered, the following disturbances have energy near $f_0$ even if the PWM ripple itself is far above it:

1. the current regulator output and reference changes, including polarity reversal of the slow-AC output;
2. the DC-link ripple: a six-pulse rectified DC link carries 300 Hz (and multiples); the bridge transfers it to the output multiplied by $m$, so with $f_0$ near 300 Hz an undamped filter would amplify it by $Q$;
3. dead-time and minimum-pulse distortion of the bridge, which produce low-frequency harmonics of the output fundamental;
4. load-current transients (load switching, crowbar or bypass operation, quench-type events);
5. PWM sidebands $2kf_{sw} \pm lf_{ac}$ at very low $f_{sw}$ combined with a resonance chosen too close to $f_{r,\min}$;

6. for a regulated downstream converter, the negative incremental input resistance of a constant-power load, which interacts with $Z_o$ (Middlebrook criterion, [1, 6, 7, 8, 9, 10]).

Active damping (a virtual resistor synthesised by the control) requires a control bandwidth several times $f_0$ and a sampling frequency well above it. With $f_{sw}$ as low as 1 kHz this is not available, so the damping has to be passive.

### 3.3 Passive damping alternatives

Fig. 3 shows the five ways of inserting one resistor into an $L$–$C$ filter; Table 3 compares them. The criteria are the DC loss (the filter carries the full load current and voltage), the preservation of the $-40$ dB/decade asymptote, the dependence of the damping on the load and the number of additional components.

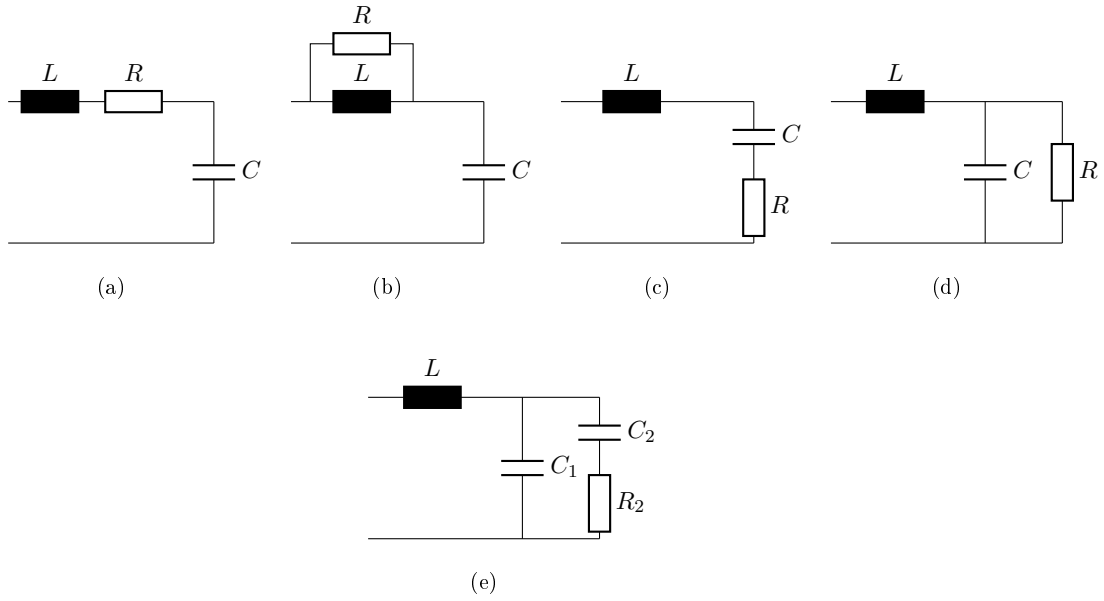



Figure 3: Single-resistor passive damping options: (a) $R$ in series with $L$; (b) $R$ in parallel with $L$; (c) $R$ in series with $C$; (d) $R$ in parallel with $C$; (e) the $C_2$–$R_2$ branch in parallel with $C_1$ considered in this note. Erickson [4] also treats (b) with a blocking inductor in series with $R$.

Option (e) is the only one that combines zero DC dissipation with the preserved $-40$ dB/dec slope and a damping that does not depend on the load. Its benefit can be summarised as follows: for a modest increase in capacitance (typically 2–4 times $C_1$, usually in a cheaper electrolytic technology since $C_2$ carries little ripple current), the resonance peak of $|H|$ drops from $Q$ (tens of dB) to $(2+n)/n$ (2–6 dB), the peak of $|Z_o|$ drops from $QR_0$ to below $R_0$, the ringing after a transient is cut to a few cycles of $f_0$, and the current regulator can be designed as if the filter were a simple low-pass with a mild resonance. The costs, quantified in Sections 4 and 5, are the volume of $C_2$, the losses in $R_2$ when the output carries slow AC or fast voltage slews, and the earlier roll-off at $f_0/\sqrt{1+n}$.

## 4 Design theory of the damping branch

### 4.1 The point independent of $R_2$

The key observation, due to Erickson [4], is that all the curves $|Z_o(j\Omega)|$ obtained for a given $n$ and any value of $r$ pass through one common point. Writing (7) as

$$|H|^2 = \frac{1+x^2}{A^2+x^2B^2}, \qquad x = nr\Omega, \quad A = 1-(1+n)\Omega^2, \quad B = 1-\Omega^2, \tag{9}$$

Table 3: Comparison of the single-resistor damping options of Fig. 3

| Option | DC / slow-AC loss | High-frequency asymptote | Load dependent? | Remarks |
|---|---|---|---|---|
| (a) $R$ in series with $L$ | $RI_o^2$: unacceptable at high current | $-40$ dB/dec preserved | no | signal-level filters only |
| (b) $R$ in parallel with $L$ | none | degraded to $-20$ dB/dec above $R/(2\pi L)$; a blocking inductor $L_b$ in series with $R$ recovers it | no | Erickson's $R_f$–$L_b$ options; a second inductor is bulky at high current |
| (c) $R$ in series with $C$ | none in DC; the ripple current flows through $R$ | degraded to $-20$ dB/dec above $1/(2\pi RC)$ | no | this is what a lossy (high-ESR) capacitor does; ripple attenuation suffers |
| (d) $R$ in parallel with $C$ | $V_o^2/R$: unacceptable | preserved | no | very low power only |
| (e) $C_2$–$R_2$ branch in parallel with $C_1$ | none in DC; slow-AC term $\propto (f_{ac}C_2V_{ac})^2R_2$ (Sec. 5.3); small at $f_r$ | preserved ($C_1$ alone) | no | one extra capacitor ($nC_1$); low-frequency corner moves to $f_0/\sqrt{1+n}$; $R_2$ may be partly the ESR of $C_2$ |

$|H|$ is independent of $x$ (hence of $r$) where $A^2 = B^2$. Since $A = B$ is impossible for $n > 0$, the condition is $A = -B$, i.e.

$$\Omega_m = \sqrt{\frac{2}{2+n}}, \qquad f_m = f_0\sqrt{\frac{2}{2+n}}, \tag{10}$$

and at this frequency

$$|H(j\Omega_m)| = \frac{1}{|B|} = \frac{2+n}{n}, \qquad \frac{|Z_o(j\Omega_m)|}{R_0} = \Omega_m\,\frac{2+n}{n} = \frac{\sqrt{2(2+n)}}{n}. \tag{11}$$

Because every curve passes through this point, no choice of $r$ can bring the peak of $|H|$ below $(2+n)/n$ nor the peak of $|Z_o|$ below $R_0\sqrt{2(2+n)}/n$. The best that $R_2$ can do is to make the peak coincide with the crossing point, and this defines the optimum. Fig. 4 and Fig. 5 show the two families of curves for $n = 4$.

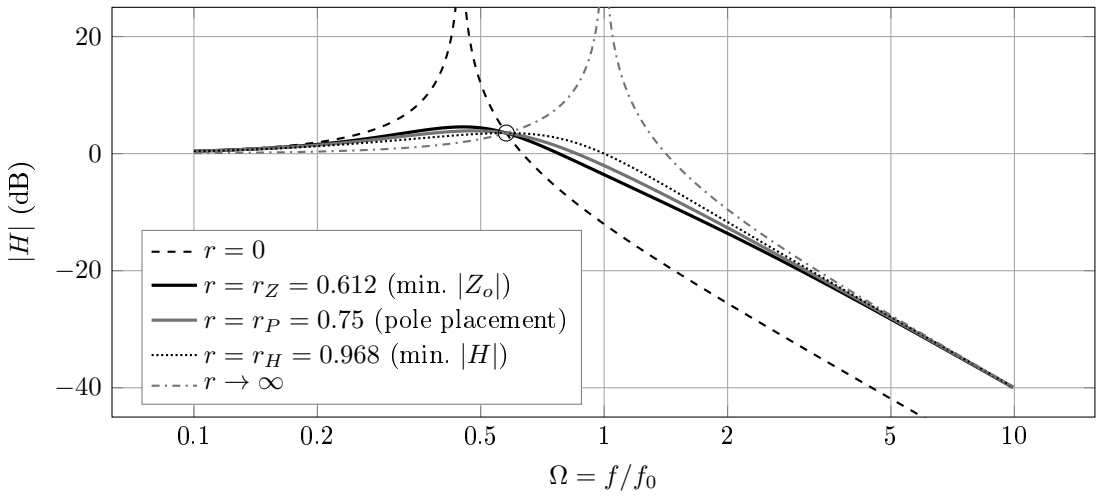



Figure 4: $|H(j\Omega)|$ for $n = 4$ and several values of $r = R_2/R_0$. All curves pass through the point $\Omega_m = \sqrt{2/(2+n)} = 0.577$, $|H| = (2+n)/n = 1.5$ (3.5 dB), marked by the circle.

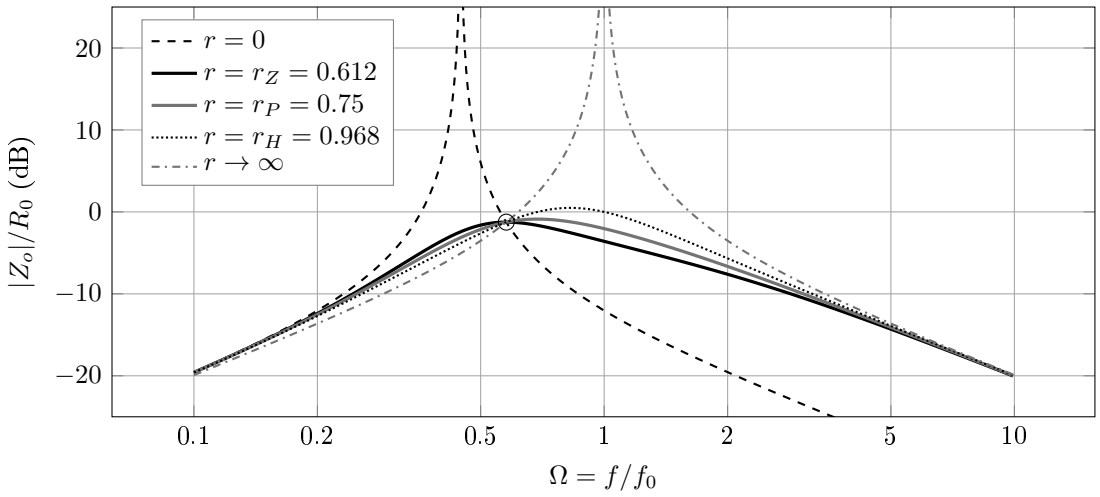



Figure 5: $|Z_o(j\Omega)|/R_0$ for $n = 4$. The common point is at $\Omega_m = 0.577$, $|Z_o|/R_0 = \sqrt{2(2+n)}/n = 0.866$ ($-1.2\,\mathrm{dB}$). The curve for $r = r_Z$ has its maximum exactly there.

### 4.2 Optimum for the output impedance (Erickson)

Imposing that the maximum of $|Z_o|$ falls at $\Omega_m$, i.e. $\partial|Z_o|^2/\partial\Omega = 0$ at $\Omega = \Omega_m$, gives the classical result [4, 5]

$$r_Z = \frac{R_{2,Z}}{R_0} = \sqrt{\frac{(2+n)(4+3n)}{2n^2(4+n)}}, \qquad \left.\frac{\|Z_o\|_\infty}{R_0}\right|_{r_Z} = \frac{\sqrt{2(2+n)}}{n}, \tag{12}$$

where $\|Z_o\|_\infty$ denotes the peak over frequency. This is the choice that minimises the peak output impedance and is the relevant one when the concern is the interaction with the load (load current steps, constant-power loads).

### 4.3 Optimum for the voltage transfer function

Since the curves of $|H|$ share the same crossing point, the same reasoning applied to $|H|^2$ (Appendix A) gives

$$r_H = \frac{R_{2,H}}{R_0} = \frac{\sqrt{(1+n)(2+n)/2}}{n}, \qquad \|H\|_\infty|_{r_H} = \frac{2+n}{n}. \tag{13}$$

This minimises the amplification of any component of the bridge voltage near the resonance and the resonant peak seen by the current regulator. $r_H$ is always larger than $r_Z$ (by a factor $\sqrt{(1+n)(4+n)/(4+3n)}$, i.e. 1.1–2 for $n = 0.5$–$8$).

### 4.4 Pole placement: real pole at $\omega_0$

A third closed-form choice follows from factorising $D(p)$. For

$$r_P = \frac{2+n}{2n} \tag{14}$$

one has $D(-1) = 2 + n - 2nr_P = 0$, so that $p = -1$ (i.e. $s = -\omega_0$) is a real pole and

$$D(p) = (1+p)\left[1 + \frac{n}{2}\,p + \frac{2+n}{2}\,p^2\right], \qquad \omega_n = \omega_0\sqrt{\frac{2}{2+n}} = \omega_0\Omega_m, \qquad \zeta = \frac{n}{2\sqrt{2(2+n)}}. \tag{15}$$

The complex pair has its natural frequency at the crossing point and a damping ratio that is close to the maximum achievable with the branch (within 7 % of it for $n \le 4$, see Appendix A). $r_P$ lies between $r_Z$ and $r_H$ for every $n$, so it is a good single compromise; the corresponding metrics are within a few percent of both optima.

### 4.5 Summary table and choice of $n$

Table 4 collects the three choices. The numbers were verified numerically (peak search over frequency and over $r$, roots of $D(p)$, step response of $H$).

Table 4: Closed-form design values of the damping branch versus $n = C_2/C_1$. $r = R_2/R_0$. $\|Z_o\|_\infty$ is given for $r_Z$, $\|H\|_\infty$ for $r_H$, $\zeta$ and the step overshoot of $v_o$ (step of $\overline{v_b}$) for $r_P$. $\Omega_m = f_m/f_0$ is the frequency of the common point, $f_{low}/f_0 = 1/\sqrt{1+n}$ is the resonance of the $L$–$(C_1+C_2)$ pair.

| $n$ | $r_Z$ | $\frac{\|Z_o\|_\infty}{R_0}$ | $r_H$ | $\|H\|_\infty$ | $r_P$ | $\zeta_P$ | OS (%) | $\Omega_m$ | $\frac{f_{low}}{f_0}$ |
|---|---|---|---|---|---|---|---|---|---|
| 0.5 | 2.472 | 4.472 | 2.739 | 5.000 | 2.500 | 0.112 | 80 | 0.894 | 0.816 |
| 1 | 1.449 | 2.449 | 1.732 | 3.000 | 1.500 | 0.204 | 68 | 0.816 | 0.707 |
| 2 | 0.913 | 1.414 | 1.225 | 2.000 | 1.000 | 0.354 | 53 | 0.707 | 0.577 |
| 3 | 0.718 | 1.054 | 1.054 | 1.667 | 0.833 | 0.474 | 44 | 0.632 | 0.500 |
| 4 | 0.612 | 0.866 | 0.968 | 1.500 | 0.750 | 0.577 | 38 | 0.577 | 0.447 |
| 6 | 0.494 | 0.667 | 0.882 | 1.333 | 0.667 | 0.750 | 30 | 0.500 | 0.378 |
| 8 | 0.427 | 0.559 | 0.839 | 1.250 | 0.625 | 0.894 | 25 | 0.447 | 0.333 |

The table shows the trade-off governed by $n$. Increasing $n$ lowers both peaks roughly as $1/n$ and raises the damping ratio, but the returns diminish above $n \approx 4$ while the capacitor volume, the low-frequency corner shift and – as shown in Section 5.3 – the losses in $R_2$ with a slow-AC output all grow with $C_2$. The usual recommendation is:

- $n = 4$ (or more) when the output is DC and the only concern is the resonance: $\|H\|_\infty \leq 3.5\,\text{dB}$, $\|Z_o\|_\infty < R_0$, $\zeta \approx 0.6$;
- $n = 2$–$3$ when the output carries slow AC or fast voltage slews, so that the reactive current $2\pi f_{ac} C_2 V_{ac}$ through $R_2$ stays acceptable: $\|H\|_\infty \leq 6\,\text{dB}$, $\zeta \approx 0.35$–$0.47$;
- $n = 1$ only when $C_2$ is very expensive (film at high voltage): the peaks are still 9.5 dB and $2.45R_0$, acceptable for many control designs but not for a demanding load interaction.

Fig. 6 shows $|Z_o|$ at the respective optimum for several $n$.

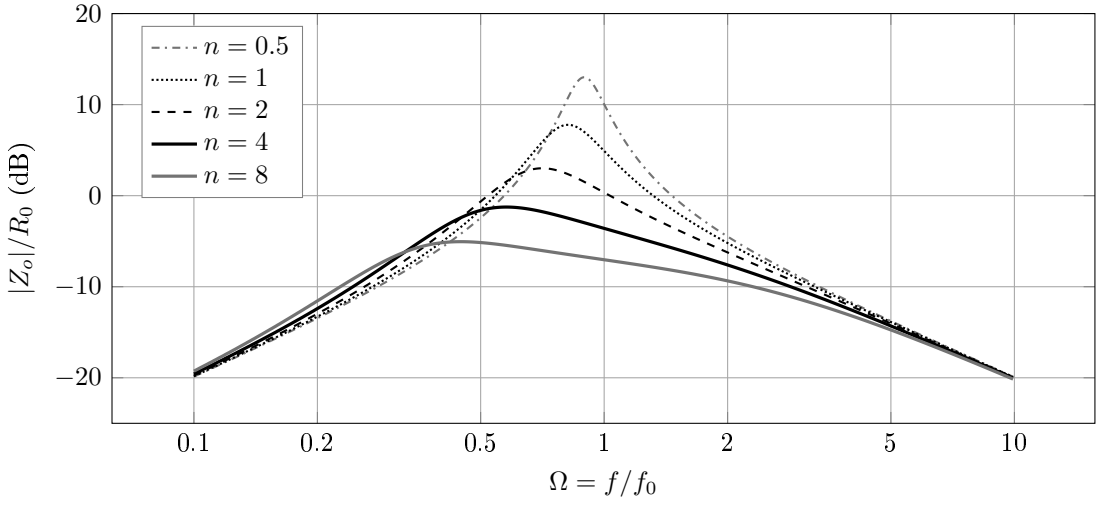



Figure 6: $|Z_o|/R_0$ with $r = r_Z$ for several capacitance ratios $n$. The peak is $\sqrt{2(2+n)}/n$ and occurs at $\Omega_m = \sqrt{2/(2+n)}$.

### 4.6 Sensitivity to the value of $R_2$

$R_2$ is never exactly at the design value: resistor tolerance, temperature coefficient, the ESR of $C_2$ (which adds to $R_2$ and changes with temperature and ageing, especially for electrolytic capacitors)

and the series resistance of the connections all contribute. Fig. 7 gives the peak of $|Z_o|$, normalised to its minimum, as a function of $R_2/R_{2,Z}$. The curve is flat around the optimum: a $\pm 25\,\%$ error costs less than $10\,\%$ (0.8 dB) of peak impedance; a factor of two costs about $50\,\%$ (3.5 dB). The design therefore tolerates the normal spread of $R_2$ and of the ESR, provided that the nominal value is chosen at the centre of the expected range and not at one extreme. Because $r_Z < r_P < r_H$, a value of $R_2$ anywhere between $r_Z R_0$ and $r_H R_0$ keeps all three metrics within a few percent of their respective optima; if the ESR of $C_2$ is uncertain it is preferable to design the external resistor so that the sum lands in this window.

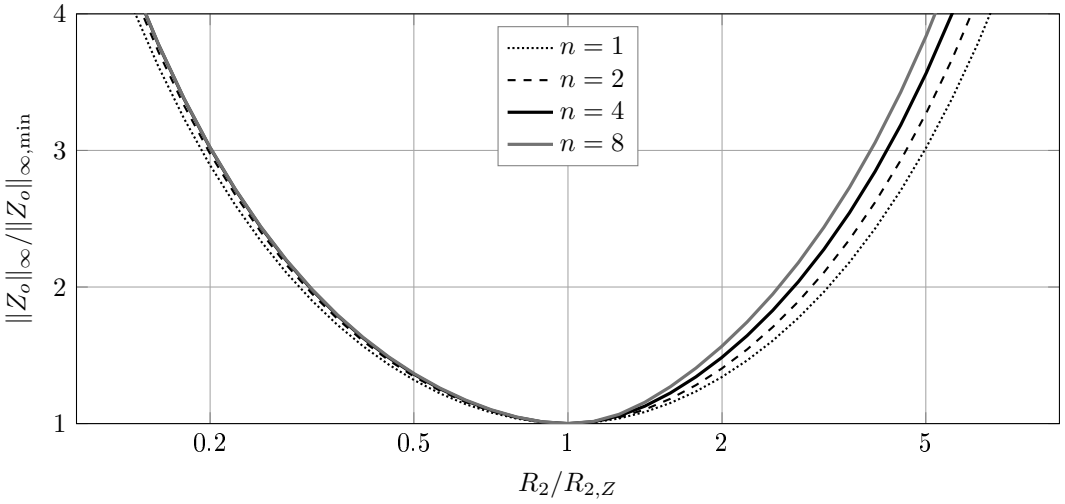



Figure 7: Sensitivity of the peak output impedance to the value of $R_2$, relative to the optimum $R_{2,Z}$, for several $n$.

### 4.7 Time-domain response

Fig. 8 shows the response of $v_o$ to a unit step of $\overline{v_b}$ for the $r_Z$ designs and for the $r_P$ design with $n = 4$; time is normalised to $1/\omega_0$. The overshoot (Table 4) is larger than the damping ratio alone would suggest, because of the zero at $\Omega_z = 1/(nr)$ in $H(s)$: the branch feeds the initial part of the step forward through $R_2$. The response settles ($2\,\%$) within about $\omega_0 t \approx 15$–$25$, i.e. two to four periods of $f_0$ for $n = 2$–$4$, against hundreds of periods for the undamped filter. In a closed-loop system the reference is rate-limited or shaped by the regulator, so this overshoot is rarely reached; the response to a load current step, governed by $Z_o$, is the more relevant transient and is bounded by $\|Z_o\|_\infty \Delta i_o$.

## 5 Design criteria for the output filter of a full bridge

### 5.1 Ripple source: unipolar PWM

With unipolar voltage switching and a modulation index $m$ that varies slowly compared with the carrier (quasi-static approximation, valid for $f_{ac} \ll f_{sw}$), the bridge output is a pulse train of amplitude $V_{dc}$, frequency $f_r = 2f_{sw}$ and duty ratio $|m|$, with average $mV_{dc}$. Its Fourier series is

$$v_b(t) = mV_{dc} + \sum_{k=1}^{\infty} \hat{V}_k \cos(2\pi k f_r t + \varphi_k), \qquad \hat{V}_k = \frac{2V_{dc}}{k\pi} \left|\sin(k\pi|m|)\right|. \tag{16}$$

The first harmonic is maximum at $|m| = 0.5$, where $\hat{V}_1 = 2V_{dc}/\pi = 0.637\,V_{dc}$ and the second harmonic vanishes; the second harmonic is maximum ($V_{dc}/\pi$) at $|m| = 0.25$ and $0.75$. For a

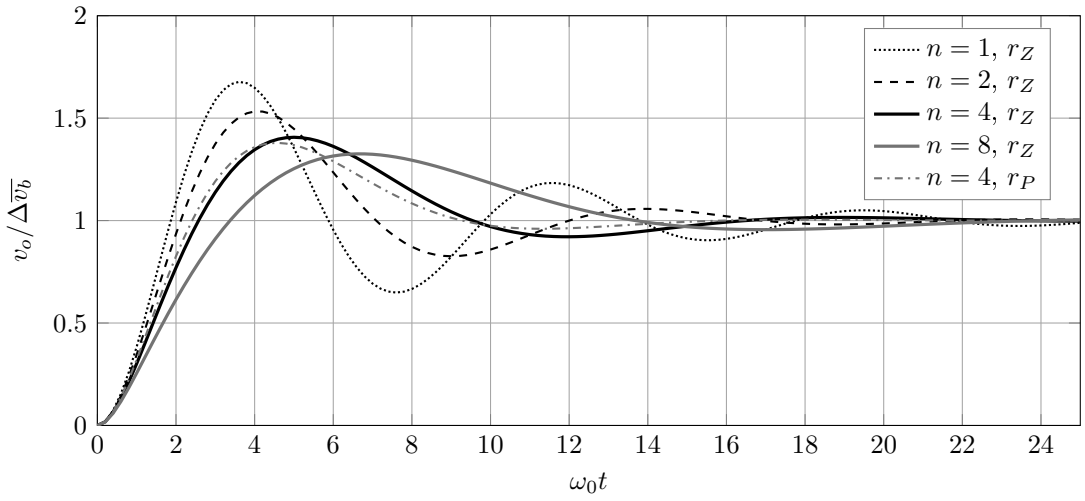



Figure 8: Response of the output voltage to a unit step of the average bridge voltage (unloaded filter), for the designs of Table 4.

sinusoidal output the same expressions hold instant by instant, so the worst case is always $\hat{V}_1 = 2V_{dc}/\pi$ at $f_{r,\min}$ [14, 15].

The inductor current ripple, with the capacitor voltage taken as constant over one ripple period, is

$$\Delta I_{L,pp} = \frac{|m|(1-|m|)\,V_{dc}}{2f_{sw}L} \le \frac{V_{dc}}{8f_{sw}L}, \tag{17}$$

triangular, with rms value $\Delta I_{L,pp}/(2\sqrt{3})$. This current flows almost entirely into $C_1$ (the branch $C_2$–$R_2$ is resistive and much larger in impedance than $C_1$ at $f_r$), so the peak-to-peak capacitor voltage ripple is

$$\Delta V_{o,pp} \simeq \frac{\Delta I_{L,pp}}{8f_r C_1} \qquad \text{or, from the first harmonic,} \qquad \Delta V_{o,pp} \simeq 2\,\hat{V}_1\,|H(j2\pi f_r)| \simeq 2\,\hat{V}_1 \left(\frac{f_0}{f_r}\right)^2. \tag{18}$$

The two estimates agree within a few percent when $f_r \ge 5f_0$.

## 5.2 Design procedure

1. **Inputs.** $V_{dc}$, rated current $I_n$, switching frequency range (worst case $f_{sw,\min}$), modulation type, output ripple specification $\Delta V_{o,pp}$ (or a load current ripple, converted through the load impedance at $f_r$), maximum output fundamental $f_{ac}$ and amplitude $\hat{V}_{ac}$, maximum voltage slew rate $(dv_o/dt)_{\max}$, load model ($L_\ell$, $R_\ell$ or a constant-power load $P$), required current-loop bandwidth $f_c$.

2. **Resonance frequency $f_0$.** From the ripple specification and (18), using the $-40$ dB/dec asymptote, which the branch does not alter,

$$f_0 \le f_{r,\min}\sqrt{\frac{\Delta V_{o,pp}}{2\hat{V}_1}} = f_{r,\min}\sqrt{\frac{\pi\,\Delta V_{o,pp}}{4V_{dc}}}. \tag{19}$$

   From the control side, the regulator crossover should satisfy $f_c \le f_0/5$ for a filter damped as in Table 4 (resonant peak $\le 6$ dB and phase lag of $H$ below about $15°$ at $f_c$). Both constraints define a window for $f_0$; with $f_{sw,\min} = 1$ kHz and a ripple of 1 % the window is roughly $f_0 \approx 120 - -180$ Hz, which caps the current-loop bandwidth at some 25–35 Hz. Keep $f_0$ (and $f_{low} = f_0/\sqrt{1+n}$) away from the DC-link ripple frequency (300 Hz for a six-pulse supply) and from $f_{ac}$ and its low harmonics.

3. **Inductor $L$.** From (17), limiting the ripple current to a fraction $\delta$ (typically 0.1–0.2) of $I_n$ at $f_{sw,\min}$, $L \geq V_{dc}/(8 f_{sw,\min}\,\delta I_n)$. Check that the reactive drop $2\pi f_{ac} L I_n$ at the output fundamental is a small fraction of $V_{dc}$ and that the core is suitable for the combination of the DC/slow-AC bias and the 2–20 kHz ripple.

4. **Capacitor $C_1$.** $C_1 = 1/(\omega_0^2 L)$; then $R_0 = \sqrt{L/C_1}$. The ripple current of $C_1$ is $\approx \Delta I_{L,pp}/(2\sqrt{3})$ at $f_r$ plus the reactive current $2\pi f_{ac} C_1 \hat{V}_{ac}/\sqrt{2}$ at the output fundamental; a film (MKP) capacitor is the natural choice, with ESL low enough to keep its series resonance above $f_{r,\max}$.

5. **Ratio $n$ and capacitor $C_2$.** Choose $n$ from Table 4 and the loss considerations of Section 5.3; $C_2 = nC_1$. $C_2$ must withstand the full output voltage (DC bias plus overshoot); its ripple current is small at $f_r$ but equal to $2\pi f_{ac} C_2 \hat{V}_{ac}/\sqrt{2}$ at the output fundamental and to $C_2\, dv_o/dt$ during slews.

6. **Resistor $R_2$.** $R_2 = rR_0$ with $r$ between $r_Z$ and $r_H$ (Table 4); $r_P$ is a good default. Subtract the ESR of $C_2$ (and of its series/parallel arrangement) from the external resistor. The resistor must be non-inductive: its inductance $L_{R2}$ should satisfy $2\pi f_{r,\max} L_{R2} \ll R_2$ (for $R_2 = 1\,\Omega$ and $f_{r,\max} = 20$ kHz this means well below 8 µH, i.e. a bifilar or thick-film construction).

7. **Load interaction.** Compare $\|Z_o\|_\infty$ with the load impedance in the band around $f_m$. For a passive $L_\ell$–$R_\ell$ load there is no stability issue; for a regulated converter drawing power $P$ at voltage $V_o$ require $\|Z_o\|_\infty \leq \frac{1}{2} V_o^2/P$ (6 dB margin on the Middlebrook criterion [1, 8]), or apply the full impedance-ratio criterion with the measured or modelled input impedance of the load.

8. **Losses and ratings** (Section 5.3).

9. **Verification.** Time-domain simulation (PLECS, Simscape) of the bridge with the real PWM, including the ESR/ESL of the capacitors and the inductor winding resistance, at $f_{sw,\min}$ and $f_{sw,\max}$, with $m = 0.5$ and with the slow-AC output at full amplitude; then measurement of $Z_o$ on the hardware with a frequency-response analyser or by a load current step.

### 5.3 Losses in the damping resistor and component ratings

The current in the branch has three origins, which add in mean square.

**Ripple at $f_r$.** The capacitor voltage ripple $\hat{V}_{o,k}$ at $kf_r$ drives the branch through the impedance $R_2 + 1/(j2\pi k f_r C_2)$, which is practically equal to $R_2$:

$$P_{R2,r} \simeq \sum_k \frac{\hat{V}_{o,k}^2}{2R_2}, \qquad \hat{V}_{o,k} \simeq \hat{V}_k\, |H(j2\pi k f_r)|. \tag{20}$$

Because $\hat{V}_{o,k}$ is already attenuated by the filter this term is small (a few watts to a few tens of watts at high power) and decreases with $f_{sw}$.

**Slow-AC output.** A sinusoidal output $\hat{V}_{ac}$ at $f_{ac}$ drives a reactive current through $C_2$, which necessarily flows in $R_2$:

$$I_{2,ac} = \frac{\hat{V}_{ac}}{\sqrt{2}} \frac{\omega_{ac} C_2}{\sqrt{1 + (\omega_{ac} R_2 C_2)^2}} \simeq \frac{\hat{V}_{ac}}{\sqrt{2}}\, \omega_{ac} C_2, \qquad P_{R2,ac} = I_{2,ac}^2 R_2. \tag{21}$$

This term grows with $f_{ac}^2$, $\hat{V}_{ac}^2$ and $C_2^2 R_2 \propto n^2 r$; it is usually the dominant loss when the output is not pure DC, and it is the reason for limiting $n$ in slow-AC applications. The same current also loads $C_2$ (ripple current rating at low frequency, where electrolytic capacitors are derated).

**Voltage slews and steps.** During a controlled slew the branch carries $i_2 \simeq C_2\, dv_o/dt$ and dissipates $R_2 i_2^2$. For an ideal step $\Delta V$ of the output voltage, the energy dissipated in $R_2$ is $\frac{1}{2} C_2 \Delta V^2$ independently of $R_2$ (the energy stored in $C_2$ is dissipated once on charge and once on discharge); with $N$ steps per second the average power is $\frac{1}{2} C_2 \Delta V^2 N$ and the resistor also needs an adequate pulse rating. In practice the slew rate of the reference should be limited so that $C_2\,(dv_o/dt)_{\max}$ stays within the continuous rating of $R_2$ and $C_2$.

**Ratings.** $R_2$: continuous power $\geq P_{R2,r} + P_{R2,ac} + P_{R2,slew}$ with margin, plus the pulse energy; low inductance. $C_1$: voltage $\geq V_{dc}$ times the overshoot factor; rms current at $f_r$ from (17). $C_2$: voltage as $C_1$; rms current from (21) and the slews; if electrolytic, series connection with balancing resistors and a check of the ESR range over temperature, since the ESR is part of $R_2$. $L$: rated for $I_n$ plus half the ripple, with the core loss at $f_r$.

## 6 Worked example

The data below are representative of the application described in Section 1.1; they are assumptions to be replaced by the actual project values in the MATLAB script of Appendix B.

Table 5: Example specification (assumed)

| | | |
|---|---|---|
| DC-link voltage | $V_{dc}$ | 800 V |
| Rated output current | $I_n$ | 500 A |
| Switching frequency per leg | $f_{sw}$ | 1–10 kHz (ripple $f_r$ 2–20 kHz) |
| Modulation | | unipolar voltage switching |
| Output | | DC or sinusoidal up to $f_{ac} = 5$ Hz, $\hat{V}_{ac} \leq 400$ V |
| Load | $L_\ell$, $R_\ell$ | 10 mH, 0.1 Ω |
| Output voltage ripple | $\Delta V_{o,pp}$ | $\leq 1\,\% V_{dc} = 8$ V at $f_{sw,\min}$, $m = 0.5$ |
| Inductor current ripple | $\delta$ | $\leq 20\,\% I_n$ at $f_{sw,\min}$ |
| Current-loop crossover | $f_c$ | $\approx 30$ Hz |

**Step 2 – $f_0$.** $\hat{V}_1 = 2 \cdot 800/\pi = 509$ V; from (19), $f_0 \leq 2000\sqrt{4/(2 \cdot 509)} = 177$ Hz. The control constraint gives $f_0 \geq 5 f_c = 150$ Hz. Target $f_0 \approx 175$ Hz.

**Step 3 – $L$.** $L \geq 800/(8 \cdot 1000 \cdot 100) = 1.0$ mH; chosen $L = 1.2$ mH, giving $\Delta I_{L,pp} = 83$ A (17 %, 24 A rms) at 1 kHz and 8.3 A at 10 kHz. The reactive drop at 5 Hz is $2\pi \cdot 5 \cdot 1.2 \cdot 10^{-3} \cdot 500 = 19$ V, negligible.

**Step 4 – $C_1$.** $C_1 = 1/(\omega_0^2 L) \rightarrow 680$ µF (film), $f_0 = 176$ Hz, $R_0 = 1.33\,\Omega$. Ripple current 24 A rms at 2 kHz; 6 A rms at 5 Hz with $\hat{V}_{ac} = 400$ V.

**Steps 5–6 – branch.** Two designs are compared in Table 6: $n = 4$ ($C_2 = 2.7$ mF) and $n = 2$ ($C_2 = 1.36$ mF), both with $r = r_P$. The undamped filter with a winding resistance of 5 mΩ is shown for reference. The load contributes no damping: $|Z_\ell(j\omega_0)| = 11\,\Omega$ and its parallel-equivalent resistance is 1.2 kΩ, i.e. $Q_{load} \approx 900$.

**Discussion.** Both damped designs meet the ripple specification (the branch does not change the attenuation at $f_r$) and reduce the resonant peak from 48 dB to 4–6 dB, so that a 30 Hz current loop can be closed with the filter treated as a mild second-order lag. The $n = 4$ design is better in every dynamic respect, but with a 400 V sinusoidal output at 5 Hz its damping resistor dissipates about 0.6 kW continuously, three times more than the $n = 2$ design, and its branch carries 24 A rms at 5 Hz, which an electrolytic $C_2$ tolerates less well than a film one. For a predominantly DC

Table 6: Worked example: results (values computed with the script of Appendix B)

| | undamped | $n = 2$, $r = r_P$ | $n = 4$, $r = r_P$ |
|---|---|---|---|
| $C_2$ | – | 1.36 mF | 2.7 mF |
| $R_2$ (incl. ESR of $C_2$) | – | 1.33 Ω | 1.0 Ω |
| $f_{low} = f_0/\sqrt{1+n}$ | – | 102 Hz | 79 Hz |
| $f_m$ | – | 125 Hz | 102 Hz |
| Complex poles | $\zeta \approx 0.002$ | 125 Hz, $\zeta = 0.35$ | 102 Hz, $\zeta = 0.58$ |
| $\|H\|_\infty$ | ≈ 266 (48 dB) | 2.06 (6.3 dB) at 114 Hz | 1.57 (3.9 dB) at 85 Hz |
| $\|Z_o\|_\infty$ | ≈ 350 Ω | 1.89 Ω at 130 Hz | 1.20 Ω at 121 Hz |
| $\|H\|$ at 2 kHz / 20 kHz | −42 / −82 dB | −42 / −82 dB | −42 / −82 dB |
| $\Delta V_{o,pp}$ at 2 kHz, $m = 0.5$ | 7.9 V | 7.9 V | 7.9 V |
| $\|H\|$, phase at 30 Hz | 1.03, 0.0° | 1.09, −1.1° | 1.14, −3.1° |
| Step overshoot / settling (2 %) | 100 % / ≈ 2 s | 53 % / 14 ms | 38 % / 12 ms |
| $I_2$, $P_{R2}$ at 2 kHz (ripple) | – | 2.1 A, 6 W | 2.8 A, 8 W |
| $I_2$, $P_{R2}$ at 5 Hz, $\hat{V}_{ac} = 165$ V | – | 5.0 A, 33 W | 9.9 A, 98 W |
| $I_2$, $P_{R2}$ at 5 Hz, $\hat{V}_{ac} = 400$ V | – | 12.1 A, 193 W | 24.1 A, 578 W |
| $I_2$, $P_{R2}$ at 5 Hz, $\hat{V}_{ac} = 600$ V | – | 18.1 A, 435 W | 36.1 A, 1300 W |
| $\hat{i}_2$ for $dv_o/dt = 18.8$ kV/s (600 V pk, 5 Hz) | – | 26 A | 51 A |

output the $n = 4$ design is preferred; for continuous slow-AC operation at large amplitude $n = 2$ (or $n = 3$) with a film $C_2$ is the better compromise, unless the loss is accepted. In both cases $R_2$ should be a non-inductive resistor of at least twice the computed continuous power, and the reference slew rate should be limited so that $C_2\, dv_o/dt$ does not exceed the rms current rating of $C_2$. Fig. 9 shows $|H|$ of the three cases against the real frequency axis together with the two ripple frequencies.

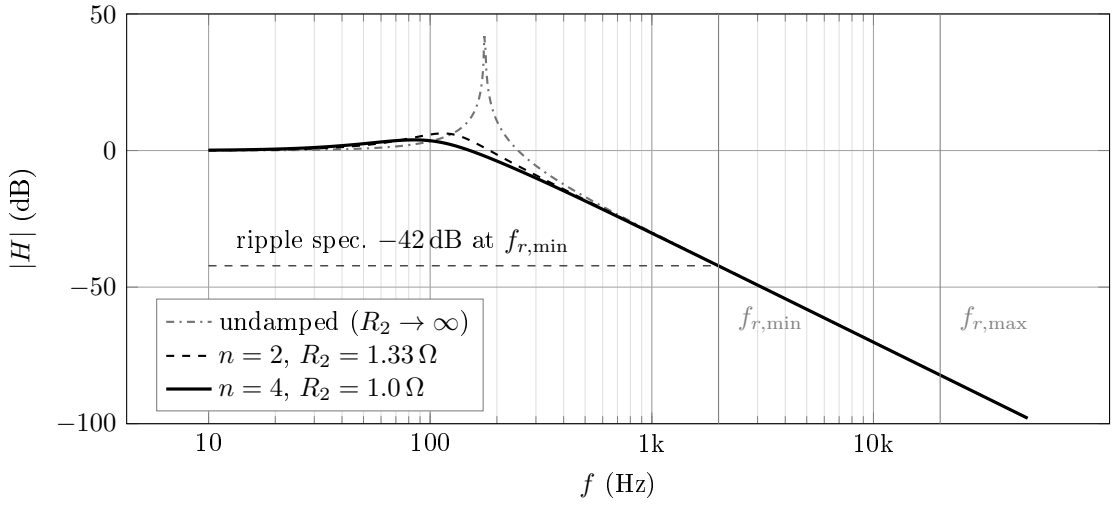



Figure 9: Worked example: $|H|$ for the undamped filter and for the two damped designs ($L = 1.2$ mH, $C_1 = 680$ µF, $f_0 = 176$ Hz). The undamped peak is limited by the parasitic resistances (here drawn without them).

## 7 Practical notes

- **Capacitor technology.** $C_1$ carries the ripple current at $f_r$ and must be a film capacitor with low ESL; $C_2$ carries little current at $f_r$ and can be an electrolytic bank (series-connected for $V_{dc} > 400$ V, with balancing resistors) when the output is DC, whereas for continuous slow-AC output a film $C_2$ avoids the low-frequency ripple-current derating of electrolytics. The ESR of $C_2$ is part of $R_2$; with electrolytics it may be a substantial part, and its range over temperature

and lifetime must be inside the flat region of Fig. 7.

- **Resistor construction.** $R_2$ must be non-inductive with respect to $f_{r,\max}$; wire-wound resistors of ordinary construction are not suitable. Thick-film on heatsink, bifilar or Ayrton-Perry wound resistors, or several resistors in parallel, are appropriate. Mount it with a thermal path sized for the slow-AC loss.
- **Parasitics.** The ESR of $C_1$ adds a zero at $1/(2\pi\,\mathrm{ESR}\,C_1)$ that flattens the $-40$ dB/dec asymptote to $-20$ dB/dec, and its ESL adds a series resonance above which the attenuation stops improving. Both must be above $f_{r,\max}$ with margin. The winding resistance of $L$ helps the damping only marginally and should not be relied upon.
- **Low-frequency corner.** With $n = 4$ the response starts to deviate from unity at $f_0/\sqrt{5}$; if the current loop or the slow-AC output needs a flat response up to a given frequency, check $|H|$ and $\arg H$ there rather than at $f_0$.
- **DC-link ripple.** The DC-link ripple appears at the output multiplied by $m$ and by $|H|$ at its frequency. If the six-pulse ripple at 300 Hz lies just above $f_0$, $|H|$ there is still only about $-10$ dB (example of Section 6); size the DC-link capacitor accordingly or move $f_0$ lower.
- **Protection.** On a crowbar or bypass event at the load, the filter capacitors discharge through the fault; $C_2$ discharges through $R_2$, which limits its contribution, while $C_1$ is limited only by the loop inductance. Consider this in the fault current budget and in the pulse rating of $R_2$.
- **Measurement.** The most direct check on the hardware is the output impedance, measured by injecting a current at the output with the bridge in freewheeling or at zero modulation, or the response of $v_o$ to a load current step; the peak and its frequency should match $\|Z_o\|_\infty$ and $f_m$ of Table 4.

# 8 Conclusions

The $C_2$–$R_2$ branch in parallel with the filter capacitor is the passive damping that best suits a high-current DC or slow-AC output filter: it dissipates nothing in DC, leaves the $-40$ dB/decade ripple attenuation untouched and damps the resonance regardless of the load. Its design reduces to two dimensionless numbers. The ratio $n = C_2/C_1$ sets the achievable peaks, $\|H\|_\infty = (2+n)/n$ and $\|Z_o\|_\infty = R_0\sqrt{2(2+n)}/n$, at the frequency $f_0\sqrt{2/(2+n)}$; $n = 2$–$4$ is the practical range. The resistance $r = R_2/R_0$ is chosen between the Erickson optimum $r_Z$ (minimum output impedance peak) and $r_H$ (minimum transfer-function peak), the pole-placement value $r_P = (2+n)/(2n)$ being a convenient default; the result is insensitive to $\pm 25\,\%$ errors in $R_2$. The cost of the branch is the capacitor $C_2$ and, when the output carries slow AC or fast slews, the reactive current $2\pi f_{ac} C_2 \hat{V}_{ac}$ that flows through $R_2$, which must be included in the loss budget and in the ratings of $R_2$ and $C_2$ and which limits $n$ in such applications.

# A Derivations

## A.1 Shunt network and transfer functions

With $Z_1 = 1/(sC_1)$ and $Z_2 = R_2 + 1/(sC_2) = (1 + sR_2C_2)/(sC_2)$,

$$Z_c = \frac{Z_1 Z_2}{Z_1 + Z_2} = \frac{\dfrac{1 + sR_2C_2}{s^2 C_1 C_2}}{\dfrac{(C_1 + C_2) + sR_2C_1C_2}{sC_1C_2}} = \frac{1 + sR_2C_2}{s\left[(C_1 + C_2) + sR_2C_1C_2\right]}, \tag{22}$$

which is (1) with $C_s = C_1C_2/(C_1+C_2)$. Then

$$H = \frac{Z_c}{sL+Z_c} = \frac{1+sR_2C_2}{s^2L\left[(C_1+C_2)+sR_2C_1C_2\right]+1+sR_2C_2}, \tag{23}$$

which is (2); $Z_o = sLZ_c/(sL+Z_c) = sL\,H$ and $Z_{in} = sL+Z_c$ follow directly. With $s = p\,\omega_0$, $\omega_0^2LC_1 = 1$, $\omega_0R_0C_1 = 1$, $C_2 = nC_1$ and $R_2 = rR_0$:

$$sR_2C_2 = p\,\omega_0\,rR_0\,nC_1 = nr\,p, \tag{24}$$

$$s^2L(C_1+C_2) = p^2\,\omega_0^2LC_1(1+n) = (1+n)\,p^2, \tag{25}$$

$$s^3LR_2C_1C_2 = p^3\,\omega_0^3LC_1\,rR_0\,nC_1 = nr\,p^3, \tag{26}$$

giving (5)–(6).

### A.2 Common point of the magnitude curves

With $x = nr\Omega$, $A = 1-(1+n)\Omega^2$, $B = 1-\Omega^2$,

$$|H|^2 = \frac{1+x^2}{A^2+x^2B^2}. \tag{27}$$

$|H|^2$ is independent of $x$ if and only if $A^2 = B^2$; $A = B$ requires $n\Omega^2 = 0$, so $A = -B$: $1-(1+n)\Omega^2 = \Omega^2-1$, i.e. $\Omega_m^2 = 2/(2+n)$. There $B = 1-2/(2+n) = n/(2+n)$ and $|H| = 1/|B| = (2+n)/n$, $|Z_o|/R_0 = \Omega_m|H| = \sqrt{2/(2+n)}\,(2+n)/n = \sqrt{2(2+n)}/n$. For any $r$ the peak of a curve is at least the value at the common point, so these are lower bounds on the achievable peaks.

### A.3 Optimum values of $r$

Let $G(\Omega) = |H|^2$ and $F(\Omega) = \Omega^2|H|^2 = |Z_o|^2/R_0^2$. A curve whose maximum lies at the common point satisfies $\partial G/\partial\Omega = 0$ (resp. $\partial F/\partial\Omega = 0$) at $\Omega = \Omega_m$. Carrying out the differentiation of the rational functions and substituting $\Omega_m^2 = 2/(2+n)$ yields a single equation in $r^2$ in each case, with the positive roots

$$r_Z^2 = \frac{6n^2+20n+16}{4n^2(n+4)} = \frac{(2+n)(4+3n)}{2n^2(4+n)}, \qquad r_H^2 = \frac{2n^2+6n+4}{4n^2} = \frac{(1+n)(2+n)}{2n^2}. \tag{28}$$

The first is Erickson's result [4]; both were checked numerically by a direct minimisation of the peak over $r$ (Section 4). Note that $r_H > r_Z$ for all $n > 0$: $r_H/r_Z = \sqrt{(1+n)(4+n)/(4+3n)}$, which grows slowly with $n$.

### A.4 Pole placement

$D(p) = 1+nr\,p+(1+n)\,p^2+nr\,p^3$ has $p = -1$ as a root when $1-nr+(1+n)-nr = 0$, i.e. $r = r_P = (2+n)/(2n)$, for which $nr_P = (2+n)/2$. Dividing by $(1+p)$,

$$D(p) = (1+p)\left(\frac{2+n}{2}\,p^2+\frac{n}{2}\,p+1\right), \tag{29}$$

so the complex pair has $\omega_n/\omega_0 = \sqrt{2/(2+n)} = \Omega_m$ and $2\zeta\,\omega_n/\omega_0 = n/(2+n)$, i.e. $\zeta = n/(2\sqrt{2(2+n)})$. Numerically the damping ratio of the complex pair as a function of $r$ has its maximum slightly above $r_P$ (0.62 against 0.58 for $n = 4$, 0.37 against 0.35 for $n = 2$).

### A.5 Unipolar PWM pulse train

For a constant modulation index $m > 0$ the leg voltages are $v_A = V_{dc}$ when the carrier is below $m$ and $v_B = V_{dc}$ when it is below $-m$; the difference $v_b = v_A - v_B$ equals $V_{dc}$ while the carrier lies between $-m$ and $m$ and is zero otherwise. With a triangular carrier this happens once in every half carrier period, for a fraction $m$ of it, so $v_b$ is a pulse train of period $1/(2f_{sw})$, amplitude $V_{dc}$ and duty ratio $m$. Its Fourier coefficients are $\hat{V}_k = (2V_{dc}/k\pi)\,|\sin(k\pi m)|$, which is (16); the result was checked by FFT of the sampled waveform for $m = 0.5$ and $m = 0.25$.

## B MATLAB script

The script below (`damped_lc_filter.m`, Control System Toolbox) reproduces the design procedure of Section 5.2 and the numbers of Section 6; the input block at the top is the only part to edit. It builds $H(s)$, $Z_o(s)$, $Z_{in}(s)$ and the loaded transfer function, prints the peaks, the poles and the loss estimates, sweeps $R_2$ around the optimum and plots the frequency and step responses in grey scale.

```
%% damped_lc_filter.m
% L - C1 - (C2 + R2) output filter of an IGBT full bridge (unipolar PWM).
% Companion of "Passively Damped LC Output Filter with R-C Damping Branch",
% Rev. 00, 2026-09-21.  Requires the Control System Toolbox.
clear; clc; close all

%% ---- Input data (edit) ------------------------------------------------
Vdc   = 800;          % DC-link voltage [V]
In    = 500;          % rated output current [A]
fsw   = [1e3 10e3];   % switching frequency range per leg [Hz]
dVpp  = 0.01*Vdc;     % output voltage ripple spec, peak-to-peak [V]
delta = 0.20;         % max inductor current ripple, fraction of In
fc    = 30;           % current-loop crossover [Hz]
fac   = 5;            % max output fundamental [Hz]
Vac   = 400;          % max slow-AC output amplitude [V peak]
dvdt  = 2*pi*fac*600; % max output voltage slew rate [V/s]
Lload = 10e-3;  Rload = 0.1;      % load model
n     = 4;            % C2/C1
crit  = 'P';          % 'Z' Erickson (min peak Zo), 'H' (min peak H), 'P' (pole at -w0)
L     = 1.2e-3;       % chosen inductor [H]   (see step 3)
C1    = 680e-6;       % chosen capacitor [F]  (see step 4)

%% ---- Step 2: resonance frequency window -------------------------------
fr    = 2*fsw;                      % ripple frequency (unipolar PWM)
V1    = 2*Vdc/pi;                   % worst-case first harmonic (m = 0.5)
f0max = min(fr)*sqrt(dVpp/(2*V1));  % from the -40 dB/dec asymptote
f0min = 5*fc;                       % from the control bandwidth
fprintf('f0 window: %.0f ... %.0f Hz\n', f0min, f0max);

%% ---- Step 3: inductor -------------------------------------------------
Lmin  = Vdc/(8*min(fsw)*delta*In);
dIpp  = Vdc./(8*fsw*L);             % worst-case ripple (m = 0.5)
fprintf('L >= %.2f mH, chosen %.2f mH -> dI_pp = %.1f / %.1f A\n', ...
        Lmin*1e3, L*1e3, dIpp);

%% ---- Step 4: capacitor and base quantities ----------------------------
w0 = 1/sqrt(L*C1);  f0 = w0/(2*pi);  R0 = sqrt(L/C1);
fprintf('f0 = %.1f Hz, R0 = %.3f ohm\n', f0, R0);

%% ---- Steps 5-6: damping branch ----------------------------------------
rZ = sqrt((2+n)*(4+3*n)/(2*n^2*(4+n)));   % Erickson, min peak |Zo|
rH = sqrt((1+n)*(2+n)/2)/n;               % min peak |H|
rP = (2+n)/(2*n);                         % real pole at -w0
switch crit
    case 'Z', r = rZ;
    case 'H', r = rH;
    otherwise, r = rP;
end
C2 = n*C1;  R2 = r*R0;
fprintf('n = %g: rZ = %.3f, rH = %.3f, rP = %.3f -> R2 = %.3f ohm, C2 = %.2f mF\n', ...
        n, rZ, rH, rP, R2, C2*1e3);
```

```
fprintf('closed form: peak|Zo| = %.3f ohm (rZ), peak|H| = %.3f (rH), fm = %.1f Hz\n', ...
        R0*sqrt(2*(2+n))/n, (2+n)/n, f0*sqrt(2/(2+n)));

%% ---- Transfer functions ------------------------------------------------
s   = tf('s');
Zc  = (1 + s*R2*C2)/(s*((C1+C2) + s*R2*C1*C2));   % shunt network
H   = minreal(Zc/(s*L + Zc));                      % vo/vb, unloaded
Zo  = minreal(s*L*H);                              % output impedance
Zin = minreal(s*L + Zc);                           % seen by the bridge
Zl  = Rload + s*Lload;
Zcl = minreal(Zc*Zl/(Zc + Zl));
Hl  = minreal(Zcl/(s*L + Zcl));                    % vo/vb, loaded
Hund = 1/(1 + s^2*L*C1);                           % undamped reference

[gH, wH] = getPeakGain(H);   [gZ, wZ] = getPeakGain(Zo);
fprintf('peak|H| = %.3f (%.1f dB) at %.0f Hz; peak|Zo| = %.3f ohm at %.0f Hz\n', ...
        gH, 20*log10(gH), wH/(2*pi), gZ, wZ/(2*pi));
damp(H)                                            % poles, zeta, frequencies

%% ---- Ripple check ------------------------------------------------------
for k = 1:numel(fr)
    Hk  = abs(freqresp(H, 2*pi*fr(k)));
    Vo1 = V1*Hk;                                   % first harmonic at output
    w   = 2*pi*fr(k);
    I2  = Vo1/sqrt(2)/abs(R2 + 1/(1j*w*C2));       % branch current, rms
    fprintf('fr = %5.0f Hz: |H| = %.1f dB, dVo_pp = %.2f V (%.2f %%), I2 = %.2f A rms, P_R2 = %.1f W\n', ...
            fr(k), 20*log10(Hk), 2*Vo1, 200*Vo1/Vdc, I2, I2^2*R2);
end
fprintf('triangular estimate: dVo_pp = %.2f V at fr,min\n', dIpp(1)/(8*fr(1)*C1));

%% ---- Slow-AC and slew losses in R2 -------------------------------------
wac  = 2*pi*fac;
I2ac = Vac/sqrt(2)*wac*C2/sqrt(1 + (wac*R2*C2)^2);
I1ac = Vac/sqrt(2)*wac*C1;
fprintf('%g Hz, %g V pk: I2 = %.1f A rms -> P_R2 = %.0f W; I_C1 = %.1f A rms\n', ...
        fac, Vac, I2ac, I2ac^2*R2, I1ac);
fprintf('slew %.1f kV/s: i2 = %.0f A pk, %.1f kW in R2 during the slew\n', ...
        dvdt/1e3, C2*dvdt, (C2*dvdt)^2*R2/1e3);

%% ---- Low-frequency behaviour (control) ---------------------------------
for f = [fac fc 2*fc]
    h = freqresp(H, 2*pi*f);
    fprintf('f = %4.0f Hz: |H| = %.3f, phase = %.1f deg\n', f, abs(h), angle(h)*180/pi);
end

%% ---- Sensitivity of the peaks to R2 ------------------------------------
kk = logspace(-1, 1, 41);  pkZ = zeros(size(kk));  pkH = pkZ;
for i = 1:numel(kk)
    R2i = kk(i)*rZ*R0;
    Zci = (1 + s*R2i*C2)/(s*((C1+C2) + s*R2i*C1*C2));
    Hi  = minreal(Zci/(s*L + Zci));
    pkH(i) = getPeakGain(Hi);  pkZ(i) = getPeakGain(minreal(s*L*Hi));
end

%% ---- Plots (grey scale) ------------------------------------------------
fv = logspace(0, 5, 2000);  w = 2*pi*fv;
figure('Name', 'Voltage transfer function');
semilogx(fv, 20*log10(squeeze(abs(freqresp(Hund, w)))), '-.', 'Color', [.55 .55 .55], 'LineWidth', 1.2);
    hold on
semilogx(fv, 20*log10(squeeze(abs(freqresp(H, w)))),    'k-',  'LineWidth', 1.6);
semilogx(fv, 20*log10(squeeze(abs(freqresp(Hl, w)))),   'k--', 'LineWidth', 1.2);
xline(fr(1), ':', 'Color', [.4 .4 .4]); xline(fr(end), ':', 'Color', [.4 .4 .4]);
grid on; ylim([-100 50]); xlabel('f (Hz)'); ylabel('|H| (dB)');
legend('undamped', 'damped, unloaded', 'damped, loaded', 'Location', 'southwest');

figure('Name', 'Output impedance');
semilogx(fv, 20*log10(squeeze(abs(freqresp(Zo, w)))), 'k-', 'LineWidth', 1.6); hold on
semilogx(fv, 20*log10(abs(1j*w*L)), ':', 'Color', [.5 .5 .5]);
semilogx(fv, 20*log10(1./(w*C1)), ':', 'Color', [.5 .5 .5]);
grid on; xlabel('f (Hz)'); ylabel('|Z_o| (dB ohm)'); ylim([-40 20]);
legend('|Z_o|', '\omega L', '1/(\omega C_1)', 'Location', 'south');

figure('Name', 'Peak sensitivity');
```

```
semilogx(kk, pkZ/min(pkZ), 'k-', kk, pkH/min(pkH), 'k--', 'LineWidth', 1.4);
grid on; xlabel('R_2 / R_{2,Z}'); ylabel('peak / minimum peak');
legend('|Z_o|', '|H|', 'Location', 'north');

figure('Name', 'Step response');
[y, t] = step(H, 0.06);  [yl, tl] = step(Hl, 0.06);
plot(t*1e3, y, 'k-', tl*1e3, yl, 'k--', 'LineWidth', 1.4); grid on
xlabel('t (ms)'); ylabel('v_o / \Delta v_b'); legend('unloaded', 'loaded');
S = stepinfo(H);
fprintf('step: overshoot %.1f %%, settling (2 %%) %.1f ms\n', S.Overshoot, S.SettlingTime*1e3);
```

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